**Biopochito: A Culturally Adapted Science Popularization Platform for Environmental Education and Green Media Literacy**

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**Abstract**

**Background:** The global environmental crisis requires educational approaches that transcend information transmission and promote systems thinking.
**Objectives:** This study evaluates the implementation of a culturally adapted Green Media and Information Literacy (Green MIL) workshop using an AI companion (Biopochito) and an interactive knowledge environment (the Living Portal).
**Methods:** A mixed-methods, exploratory case study was conducted at a school in La Paz, Bolivia (3,600 meters above sea level). Fifty-two children (aged 8 to 17, divided into three age-specific groups: 8–10, 11–14, and 15–17) and their families participated in a 4-hour STEAM workshop across three stations. The study integrated quantitative methods (pre/post surveys) with qualitative methods (non-participant observation and artifact rubrics). Due to non-normal distributions, a Wilcoxon signed-rank test was employed.
**Results:** After Bonferroni correction, the analysis revealed statistically significant short-term improvements in Knowledge ($p < 0.0001$, $r = 0.78$), Environmental Attitude ($p = 0.0053$, $r = 0.52$), and Curiosity ($p = 0.0077$, $r = 0.48$). MIL Self-Efficacy showed a nominal pre–post improvement that did not remain statistically significant after correction ($p = 0.0477$, $r = 0.33$). Qualitative analysis of student artifacts confirmed the impact of the "Arts" component in STEAM, with video productions scoring an average of 3.42 out of 4.0 for responsible communication and empathy.
**Conclusions:** The findings provide preliminary evidence of positive short-term changes following participation. Due to the exploratory design and lack of a control group, the specific effect of the AI component cannot be isolated from the broader educational context.
**Implications:** Culturally adapted AI tools may support science education in similar exploratory contexts, though controlled and longitudinal studies are needed to confirm these results.

**Keywords:** Media and Information Literacy; Green MIL; Science Culture Construction; Educational Artificial Intelligence; Environmental Education.

**Practitioner Notes**

**What is already known about this topic:**

- Media and Information Literacy (MIL) is crucial for navigating climate misinformation, but traditional MIL education often remains abstract and policy-focused.
- Artificial Intelligence (AI) is increasingly being integrated into educational settings, yet there is a scarcity of documented AI implementations in Latin American school contexts, particularly those integrating local ecological knowledge.
- Systems thinking and environmental attitudes are difficult to teach through traditional didactic methods.

**What this paper adds:**

- A practical, culturally adapted implementation model (the Biopochito platform) that integrates AI, ancestral knowledge, and Green MIL in a vulnerable, high-altitude context.
- Empirical evidence showing significant short-term improvements in children's environmental knowledge, attitudes, and curiosity following a 4-hour STEAM workshop.
- Evidence that creative artifact production (stories, art, and videos) can successfully translate scientific knowledge into empathetic and responsible environmental communication.

**Implications for practice and/or policy:**

- Educators in the Global South can use culturally adapted, low-energy AI companions (like Biopochito) to foster curiosity and environmental awareness without requiring high-end infrastructure.
- Policymakers should consider supporting interdisciplinary educational models that bridge scientific evidence, local cultural knowledge, and digital literacy.
- Future educational interventions should plan for longitudinal follow-ups and explore implementation in rural contexts to democratize scientific knowledge.

## 1. Introduction

### 1.1 Background and Context

The Anthropocene is defined by interconnected crises—climate change, biodiversity loss, and pollution—that transcend national boundaries and disciplinary silos (Steffen et al., 2015). Addressing these crises requires systems thinking: the ability to recognise feedback loops and interdependencies among ecological, technological, and social subsystems (Meadows, 2008).

However, systems thinking is notoriously difficult to teach, particularly in hard-science domains where students are trained to isolate variables (Hmelo-Silver & Azevedo, 2006).

In this context, science communication and STEAM education (Science, Technology, Engineering, Arts, and Mathematics) emerge as fundamental strategies for building scientific culture (Bucchi & Trench, 2014; Miller, 2004). The integration of the Arts is particularly relevant, as creativity and artistic expression are essential for connecting critical thinking, communication, and science.

UNESCO's Green Media and Information Literacy (Green MIL) framework emphasises the need to verify environmental claims, assess information sources, and engage critically with scientific evidence (UNESCO, 2021). Science Culture Construction (SCC), a paradigm developed by Andean Road Countries for Science and Technology (ARCST), seeks to integrate science, education, culture, and technology into everyday life (Cabero, 2023). However, there is a scarcity of documented implementation experiences in Latin American school contexts, particularly those integrating Artificial Intelligence (AI) and local knowledge.

Recent studies have highlighted the potential of generative AI to enhance environmental education by providing personalised and interactive learning experiences (Cooper, 2023; Zhai et al., 2024). However, concerns remain regarding AI hallucinations and the ethical use of these tools with minors. Biopochito addresses these concerns through its localised architecture and pedagogical design, which prioritises critical thinking and responsible communication over mere information retrieval.

### 1.2 The Biopochito Platform

Biopochito originated in Colombia through the collaboration between the Unidad Central del Valle del Cauca (UCEVA) and the International Green Science Center for Latin America and the Caribbean (IGSCLAC), as documented in the recent volume on China–Latin America cooperation for green development (Cunhai & Pochmann, 2025). In 2025, ARCST transformed Biopochito into an AI-powered companion.

Technically, Biopochito is an AI companion deployed on a custom, compact hardware device (dimensions: 4 × 4.5 × 2 cm). The device connects to the internet to process natural language and generate responses. To ensure safety and pedagogical alignment, the system uses a rule-based Retrieval-Augmented Generation (RAG) architecture. It primarily retrieves facts from a curated, cloud-hosted database of approximately 500 text chunks focused on Lake Titicaca. While the device has internet access, the RAG architecture ensures that the model grounds its answers in the verified educational database rather than relying solely on the open web, thereby minimizing hallucinations.

To validate AI-generated responses and reduce hallucinations, the Biopochito system was designed with a strict retrieval threshold: if a query falls outside the verified database parameters, the model is instructed to defer with a pedagogical prompt: *"That is a great question! What do you think we could do to find out?"* Furthermore, facilitators monitored all interactions in real-time, and a post-session audit was conducted on a random sample of 20% of the dialogues to

ensure factual accuracy. No personal data, audio, or student identifiers were stored on the Biopochito devices or the tablets during the workshop.

### 1.3 Research Questions

This study aims to address the following research questions (RQs):
RQ1: How was the Green MIL activity implemented at Unidad Educativa Sariri?
RQ2: What were the short-term changes in curiosity, MIL self-efficacy, environmental attitudes, and knowledge following the workshop?
RQ3: What qualitative evidence of engagement, critical questioning, and MIL competencies emerged during the activities?

1.4 Specific Contributions of this Study
To clarify the novelty of this work beyond the combination of existing elements, we articulate our contributions as follows: (1) Technical: We document an offline Retrieval-Augmented Generation (RAG) architecture (Biopochito) designed specifically for child-oriented science education without internet access. (2) Pedagogical: We present a Green MIL design aimed at environmental misinformation and verification. (3) Empirical: We provide a feasibility and implementation study in a Latin American school, detailing short-term changes. (4) Cultural: We describe a culturally adapted interaction model grounded in local ecological knowledge. (5) Ethical: We outline a reproducible safety and data-minimisation protocol for AI use with minors. We do not present this prototype as a validated educational technology, but as an exploratory implementation. Regarding cultural adaptation, the local knowledge (e.g., Lake Titicaca biodiversity, totora reeds) was selected through consultation with local educators at Unidad Educativa Sariri. These educators validated the accuracy and age-appropriateness of the content, ensuring respectful representation. No Indigenous community knowledge was used without explicit educational contextualization, and the database is treated as educational content rather than proprietary data.

## 2. Methods

### 2.1 Study Design

A mixed-methods, exploratory case study design was employed, following the TREND checklist. This design integrated qualitative methods (non-participant observation, artifact analysis, and thematic coding) with quantitative methods (pre/post surveys). The workshop was delivered twice: once on October 19, 2026, and once on October 20, 2026. The same protocol, facilitators, and observers were used in both sessions. The pre/post instruments were administered once per group. Results were pooled because the intervention and instruments were identical across both days.

Logic Model: The intervention is a multi-component educational package. We hypothesize that Station A (AI dialogue) primarily supports Knowledge and Curiosity by providing immediate,

interactive answers. Station B (Stories and Art) primarily supports Environmental Attitude by fostering emotional connection. Station C (Video) primarily supports MIL Self-Efficacy and Responsible Communication through group co-creation. The pre/post survey measures the combined effect of these components. However, this exploratory study does not isolate the specific contribution of each station.

### 2.2 Participants and Participant Flow

Fifty-three students enrolled in the activity. However, the intervention was specifically designed to accommodate three distinct age groups to tailor the pedagogical approach to their developmental stages: Group 1 (ages 8–10, n=14), Group 2 (ages 11–14, n=18), and Group 3 (ages 15–17, n=20). All groups participated in the same three-station workshop, but facilitators adapted the complexity of the AI interactions and the depth of the creative activities according to the age group.

Fifty-two students completed both pre- and post-assessments and were included in the paired analysis. One participant was excluded because of incomplete post-test data. Sampling was non-probabilistic and by convenience. The school authorised the study, and written informed consent was obtained from parents/guardians, with oral assent from the children. All four outcome measures had 52 complete pre/post pairs.

### 2.3 Intervention Timeline

The 4-hour intervention (per session) followed this structure:

| Activity | Duration |
|---|---|
| Registration and Orientation | 20 min |
| Pre-Test | 15 min |
| Station A (AI Dialogue) | 45 min |
| Station B (Stories and Art) | 45 min |
| Break | 15 min |
| Station C (Video & Family Co-Creation) | 45 min |

| Activity | Duration |
|---|---|
| Plenary | 30 min |
| Post-Test | 15 min |
| Closing | 10 min |

All children experienced all three stations.

### 2.4 The Living Portal

The Living Portal is a web-based repository accessed via tablets connected to the school's local network. It was used primarily in Station B and the Plenary to display images of Lake Titicaca's biodiversity (e.g., *Telmatobius culeus*, *Orestias* fish). The portal also provided a secure platform for families to upload and share their recorded videos. To ensure data privacy, video uploads were restricted to the school's private network and permanently deleted from the portal immediately after the workshop. Temporary files were cleared from browser caches, and no backups were retained. Transcripts were anonymised. Families and guardians consented to the recording and sharing of videos. The portal was designed to not store metadata or record user activity.

*Note on hardware integration:* The AI interactions were managed independently by the compact **Biopochito hardware device**, which operated alongside the tablets. The tablets served only as a visual interface for the Living Portal and to display the AI responses generated by the Biopochito device.

### 2.5 Instruments and Data Collection

**Quantitative (Pre/Post Survey):** A 5-section instrument designed by the authors. The survey was administered in Spanish, adapted to the local context. Scores were calculated by summing the items.

- **Section A (Curiosity):** 5 items, Likert 1–5. Range: 5–25.
- **Section B (MIL Self-Efficacy):** 8 items, Likert 1–5. Range: 8–40.
- **Section C (Environmental Attitude):** 5 items, Likert 1–5. Range: 5–25.
- **Section D (Knowledge):** 4 multiple-choice + 4 open-ended items. Range: 0–16.
- **Section E (Open-ended):** 2 qualitative questions.

The pre/post survey was subjected to expert review by three educators and one psychometrician to ensure content validity and age-appropriate language. Additionally, a pilot test was conducted with 10 students of similar age prior to the intervention to assess item comprehension. While

formal reliability assessment (e.g., Cronbach's alpha) was not feasible due to the exploratory nature of the study and the small sample size, the pilot test confirmed the clarity of the items and the internal consistency of the constructs.

Qualitative (Observation and Artifacts): The study employed non-participant observation as the primary qualitative technique. Descriptive observation sheets were used to systematically record children's interactions, questions, and engagement during the three stations. Additionally, responses to Section E were collected. The analysis of qualitative results was based on the content of the stories, drawings, and videos produced by the children, which were evaluated using a 4-point rubric. This dual approach allowed for methodological triangulation, strengthening the validity of the findings by comparing self-reported survey data with observed behaviors and creative artifacts.

To ensure construct validity for our Green MIL outcomes, we mapped each claimed competency to observable behaviors: (a) Curiosity was measured by the pre/post survey and coded from open-ended questions; (b) Verification (MIL) was mapped to items asking 'How do we know?', scored on the E1 rubric; (c) Responsible Communication was mapped to the Station C video rubric criteria (Evidence, Empathy, Call to Action). Environmental advocacy statements were not treated as source verification, but as a separate outcome of responsible communication.

### 2.6 Data Analysis

Effect sizes were calculated using the signed rank-biserial correlation (r). Medians and interquartile ranges (IQR) were used as descriptive statistics. The 95% confidence intervals (CI) for the median differences were calculated using a bootstrap approach (2000 resamples). The number of positive, negative, and zero differences were also recorded.

Qualitative analysis involved thematic coding of the observation sheets and open-ended responses, as well as a content analysis of the children's artifacts (stories, drawings, and videos) using the established rubric. The qualitative themes were derived directly from the post-test rubric scores of items D5, D6, D8, and E1, ensuring a reproducible and verifiable classification. All analysis scripts and anonymised data are available in the project repository.

### 2.7 Ethical Considerations

This study was conducted in accordance with the ethical principles of the Declaration of Helsinki and applicable Bolivian regulations on research with human participants and on the protection of minors' data. The study was authorised by the participating school, Unidad Educativa Sariri (La Paz, Bolivia), which approved access to the school, student participation, and the audio-visual recording procedure through Institutional Authorization Letter signed by the School Principal on October 16, 2026. Written informed consent was obtained from all parents/guardians, and oral assent was obtained from the children.

The study did not undergo review by a formally constituted research ethics committee, because no committee with jurisdiction over the study was available to the authors within the project timeline. No institutional ethical approval is therefore claimed or attributed to any institution. To mitigate this limitation, the authors (a) obtained explicit school authorisation and written parental consent plus child assent; (b) applied the data-minimisation and safety protocols described in Section 1.2 and Appendix E (no personal data or audio stored on the tablets; real-time facilitator monitoring; post-session audit of a random 20% of dialogues); and (c) adhered to the anonymisation and deletion procedures described below.

Video recordings were temporarily stored on a separate password-protected device, manually transcribed, and deleted after transcription. The device was securely cleared. No images or voices of minors are included in this publication. Anonymised data will be shared only in accordance with the consent obtained and subject to institutional restrictions. The authors acknowledge that the absence of formal ethics committee review is a limitation of this study (see Section 4.2).

### 2.8 Ancestral and Local Knowledge

The workshop incorporated local ecological knowledge about Lake Titicaca (e.g., the use of totora reeds) following consultation with local educators. The activity recognised local ecological knowledge as culturally and experientially valuable while distinguishing it from scientific claims requiring empirical validation.

## 3. Results

### 3.1 Implementation (RQ1)

The activity was successfully implemented across two sessions. 52 valid pre/post pairs were collected. The fidelity checklist indicated that the core components of the three stations were delivered as planned in both sessions.

Exploratory Analysis by Age Group: Although the study was not powered to detect subgroup differences, descriptive statistics were examined by age group (Group 1: 8–10y, n=14; Group 2: 11–14y, n=18; Group 3: 15–17y, n=20). All three groups showed positive shifts in medians across most constructs. For Knowledge, the medians increased from 3.0 to 5.0 in Group 1, from 6.5 to 8.0 in Group 2, and from 6.0 to 8.0 in Group 3. Environmental Attitude increased in all groups (G1: 23.0 to 24.0; G2: 20.0 to 21.0; G3: 20.0 to 22.0). Curiosity remained stable in Group 2 (18.5 to 18.5) but increased in Group 1 (23.5 to 24.0) and Group 3 (20.0 to 21.0). MIL Self-Efficacy showed mixed results (G1: 32.0 to 32.5; G2: 28.5 to 27.0; G3: 29.5 to 30.0). Due to developmental heterogeneity and the small sample size per cell, no significance testing was performed on these subgroups. Facilitators adapted the complexity of AI interactions for older groups, which may have influenced the depth of responses.

### 3.2 Quantitative Outcomes (RQ2)

Because the assumption of normality was violated for all difference scores (Shapiro-Wilk $p < 0.05$), the non-parametric Wilcoxon signed-rank test was used. Table 1 presents the pre- and post-test results using medians and interquartile ranges (IQR). The Wilcoxon signed-rank test revealed statistically significant short-term improvements after Bonferroni correction ($p < 0.0125$) in Knowledge ($W = 113.0$, $p < 0.0001$, $r = +0.78$), Environmental Attitude ($W = 168.0$, $p = 0.0053$, $r = +0.52$), and Curiosity ($W = 214.0$, $p = 0.0077$, $r = +0.48$). For these three constructs, the number of participants who improved (+), worsened (-), or showed no change (0) were 38/7/7, 27/10/15, and 29/11/12, respectively. MIL Self-Efficacy showed a positive nominal change ($W = 396.0$, $p = 0.0477$, $r = +0.33$), with 30 participants improving, 18 worsening, and 4 remaining unchanged; however, this result did not meet the corrected significance threshold. The 95% confidence intervals for the median differences are detailed in Table 1.

**Table 1. Pre/Post Intervention Results (N=52)**

| Construct | Pre Median (IQR) | Post Median (IQR) | Median Diff [95% CI] | + / - / 0 | W (W+ / W-) | p-value & Effect size |
|---|---|---|---|---|---|---|
| Knowledge | 6.0 (3.2) | 8.0 (3.0) | +2.0 [1.0, 3.0] | 38 / 7 / 7 | 113.0 (922.0 / 113.0) | < 0.0001* \| r = +0.78 |
| Environmental Attitude | 21.0 (5.2) | 22.0 (4.2) | +1.0 [0.0, 1.5] | 27 / 10 / 15 | 168.0 (535.0 / 168.0) | 0.0053* \| r = +0.52 |
| Curiosity | 20.0 (5.5) | 21.0 (5.0) | +1.0 [0.0, 1.0] | 29 / 11 / 12 | 214.0 (606.0 / 214.0) | 0.0077* \| r = +0.48 |
| | 30.0 (7.0) | 30.0 (7.2) | 0.0 [0.0, 2.0] | 30 / 18 / 4 | 396.0 (780.0 / 396.0) | 0.0477 \| r = +0.33 |

Note: Values are presented as medians with interquartile ranges (IQR). Differences are calculated as Post minus Pre. The + / - / 0 column indicates the number of participants who improved, worsened, or showed no change, respectively. W represents the Wilcoxon signed-rank statistic, where W+ and W- indicate the sum of positive and negative ranks. Effect size r is the signed rank-biserial correlation, defined as (W+ - W-) / (W+ + W-). Zeros were discarded and ties were handled asymptotically (SciPy 1.13). *Significant after Bonferroni correction for the four primary outcomes (threshold p = 0.0125). 95% CI = Confidence Interval for the median difference (bootstrap, 2000 resamples).

The results indicate statistically significant short-term improvements in Knowledge, Environmental Attitude, and Curiosity after Bonferroni correction. MIL Self-Efficacy showed a positive nominal change, but this result did not meet the corrected significance threshold.

### 3.3 Qualitative Outcomes: Artifacts and Communication (RQ3)

3.3 Exploratory Analyses: Artifacts and Qualitative Themes (RQ3)
Deductive Rubric Classification: The qualitative themes presented in Table 2 were derived via a deductive classification system based on the post-test rubric scores for items D5, D6, D8, and E1, rather than an inductive thematic analysis. This structured classification ensures reproducibility. Two independent coders applied the rubric, achieving 85% initial agreement (34/40 excerpts) before consensus.
Artifact Rubric Analysis: Station B (Stories and Art) artifacts were produced individually by the children. A total of 30 individual artifacts were scored by two facilitators using a 4-point rubric (dimensions: Connection, Creativity, Expression, Detail). Inter-rater reliability was confirmed by consensus scoring. The overall average score was 2.87 out of 4.0. Station C (Video) artifacts were group productions, resulting in three final videos. Therefore, the unit of analysis for Station C is the group, not the individual child. The three videos were evaluated using a 4-point rubric (dimensions: Evidence, Empathy, Call to Action, MIL), yielding an average score of 3.42 out of 4.0. Because this is based on three group products, it should be interpreted as a descriptive finding of group collaboration rather than a robust estimate of individual performance. The full rubric and evaluator guide are provided in the public repository.

**Table 2. Qualitative Themes from Open-Ended Questions**

| Theme | Example Quote (Translated) | Participants (n) | Percentage (n=52) |
|---|---|---|---|
| Initial Curiosity | "Why is the frog special?" | 18 | 34.6% |
| Causal Questioning | "What happens if the water gets dirty?" | 43 | 82.7% |
| MIL Verification | "How do we know the lake is healthy?" | 17 | 32.7% |
| Systems Thinking (Interpretive) | "If we throw trash, the fish get sick, and then we can't eat them." | 22 | 42.3% |

Note: Values are presented as medians with interquartile ranges (IQR). Differences are calculated as Post minus Pre. The + / - / 0 column indicates the number of participants who improved, worsened, or showed no change, respectively. W represents the Wilcoxon signed-rank statistic, where W+ and W- indicate the sum of positive and negative ranks. Effect size r is the signed rank-biserial correlation, defined as (W+ - W-) / (W+ + W-). Zeros were discarded and ties were handled asymptotically (SciPy 1.13). *Significant after Bonferroni correction for the four primary outcomes (threshold p = 0.0125). 95% CI = Confidence Interval for the median difference (bootstrap, 2000 resamples).

**Station C (Video and Communication):** Three group audio-visual productions were evaluated, achieving an outstanding overall average of 3.42 out of 4.0. The children chose to focus on specific local topics, including the use of totora reeds for boat construction, the Andean vizcacha, the giant Titicaca frog, and the importance of not polluting the lake to preserve its biodiversity. Regarding Media and Information Literacy (MIL) competencies, the videos demonstrated a strong grasp of responsible communication. Notable quotes included: "Caring for the environment is not an option, it is an obligation", "Do not forget to take care of the environment", and "Do not pollute Lake Titicaca." These statements reflect a shift from simple curiosity to a clear call to action and environmental stewardship.

## 4. Discussion

### 4.1 Interpretation of Findings

This study provides preliminary evidence of positive short-term changes following participation in a culturally adapted Green MIL workshop. The significant improvement in Knowledge ($p < 0.0001$, $r = 0.78$) is promising, aligning with the premise that interactive, game-based learning can facilitate retention (Gris & Bengtson, 2021). However, the exploratory design does not permit attributing these gains specifically to the AI component. The changes may have been influenced by the novelty of the activity, the participation of families, the artistic and storytelling activities, or the pre-test/post-test design itself.

The nominal pre–post improvement in MIL Self-Efficacy ($p = 0.0477$, $r = 0.33$) was not statistically significant after Bonferroni correction. Although the effect was positive and small, it should be interpreted as exploratory. This suggests that while children practiced asking "How do we know?", building the confidence to autonomously verify information requires sustained, longitudinal practice (UNESCO, 2021).

The high scores obtained in the artifact rubrics (especially in Station C) support the quantitative findings and suggest that the workshop successfully translated scientific knowledge into empathetic, responsible, and creative communication.

### 4.2 Limitations

This study has several significant limitations:

- **Lack of control group:** We cannot establish causality.
- **Sample:** The sample is small (N=52) and restricted to one school, limiting generalisability.
- **Immediate measurement:** Long-term retention was not assessed.

- AI isolation: The specific effect of the AI cannot be disentangled from the pedagogical and social context.
  Artifact units of analysis: While Station B artifacts were individual, Station C videos were group productions (n=3). The high scores for Station C reflect group dynamics and cannot be generalized to individual student competencies.
- **Absence of formal ethics committee review:** The study relied on school authorisation, written parental consent, child assent, and data-minimisation and safety protocols. Although these safeguards are consistent with the Declaration of Helsinki and applicable Bolivian regulations, the lack of formal committee review limits the external ethical oversight of the study.

### 4.3 Implications for Future Research

Future studies should employ randomised controlled trials to isolate the effect of the AI companion. Longitudinal designs are needed to assess the durability of MIL self-efficacy. Furthermore, given the positive results obtained with children in urban areas, it is highly important to apply this methodology in rural contexts. In these settings, where access to technology is limited, culturally adapted AI tools like Biopochito not only offer greater utility but also become a valuable opportunity to democratize scientific knowledge and bridge educational gaps. Additionally, adapting the Biopochito methodology for children with disabilities presents a promising avenue for inclusive AI education.

## 5. Conclusion

The implementation of Biopochito AI and the Living Portal at Unidad Educativa Sariri offers a practical example of how culturally adapted AI can be integrated into STEAM education. The study provides preliminary evidence of short-term improvements in knowledge, environmental attitudes, and curiosity. MIL Self-Efficacy showed a positive but statistically inconclusive change after correction for multiple comparisons. However, the findings should be interpreted with caution due to the exploratory design and lack of a control group. The high scores obtained in the artifact rubrics (specifically in Station C) suggest that the integration of Arts and communication successfully translated scientific knowledge into empathetic and responsible environmental action. The study suggests that culturally adapted AI tools may support science education in similar exploratory contexts, although controlled and longitudinal studies are needed to confirm these results and to determine whether the observed changes can be attributed specifically to the AI intervention or to the broader educational context.

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## Conflict of Interest Statement

The authors declare no conflict of interest. The corresponding author is not affiliated with the editorial board of the target journal.

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Declaration of Generative AI and AI-assisted technologies in the writing process: During the preparation of this work, the author(s) used ChatGPT (OpenAI) to assist with statistical code verification, English language editing, and manuscript formatting. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article. The Biopochito system itself is the subject of the study and was not used to draft this manuscript.

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## Data Availability Statement

The anonymised dataset, the Python analysis code reproducing all tables and figures, the survey instrument, the observation protocol, the fidelity checklist, the AI safety protocol, and the artifact rubrics are publicly available in the project repository: **https://github.com/12ak40/biopochito-sariri-paper**. The repository includes a README with execution instructions and documents the correspondence between the analysis scripts and the results reported in Sections 3.2 and 3.3. The consent forms and the school authorisation letter (Appendix D) contain personal and institutional information and are therefore not included in the public repository; they are submitted to the editorial office for review only and will not be made publicly available. As stated in Section 2.7, the study did not undergo formal ethics committee review, and no institutional ethical approval is claimed or attributed to any institution; data sharing is therefore governed by the consent obtained and by applicable Bolivian data-protection norms.

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**Supplementary Materials**

All supplementary materials supporting this manuscript are publicly available in the project repository: **https://github.com/12ak40/biopochito-sariri-paper**. The repository contains:

- **Appendix A:** Full Pre/Post Survey Instrument (Spanish) and Rubric Key.
- **Appendix B:** Non-participant Observation Protocol.
- **Appendix C:** Fidelity Checklist.
- **Appendix D:** Informed Consent and Assent Forms, and school authorisation (editorial review only).

- **Appendix E:** Biopochito AI Prompt Design and Safety Protocols.
- **Appendix F:** Anonymised dataset (datos.txt), Python analysis codes (verificacion_tabla1.py, verificacion_rubricas.py, verificacion_tabla2.py), and a README with execution instructions.
- **Appendix G:** Artifact Rubrics and Evaluator Guide for Stations B and C.